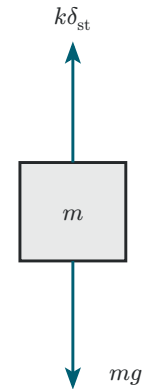
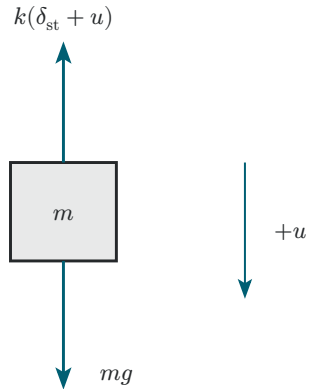


(a) Static equilibrium



$$mg - k\delta_{\text{st}} = 0$$

(b) Displaced by u



$$m\ddot{u} = -ku$$